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Date of Birth: 9th August 1983; **Gender:** M; **Category:** OBC

Academic Qualification (Undergraduate Onwards)

S. No	Degree	Year	Subject	University/ Institution
1.	M.B.B.S	2007	Medicine & Surgery	TN Govt. Dr. MGR Medical University
2.	M.Tech (University Rank holder)	2010	Nanomedical Science	Amrita Vishwa Vidhyapeetam
3.	Ph.D. (Best Thesis Awardee)	2015	Biomedical Engineering	IIT Bombay

Work Experience (in chronological order).

S. No.	Positions held	Name of the Institute	Duration
1.	CRRI (House Surgeon)	Thanjavur Medical College	Mar 2006 - Mar 2007
2.	Institute Teaching Assistant	IIT Bombay, India	July 2010 - Apr 2013
3.	Teaching Assistant under Project	IIT Bombay, India	May 2013 – Oct 2014
4.	Research Associate	IIT Bombay, India	Nov 2014 - July 2015
5.	Assistant Professor	IIT Hyderabad, India	Aug 2015 – Oct 2021
6.	Associate Professor	IIT Hyderabad, India	Oct 2021 – Till date

Professional Recognition/ Award/ Fellowship (Selected)

S. No	Name of Award	Awarding Agency	Year
1	Institute Teaching Excellence Award	IIT Hyderabad	2024
2	G D NAIDU Young Scientist Award	People's Thinking Forum -TN	2023
3	MERK Young Scientist Award	MERK Science Foundation	2023
4	SERB – Biomedical -PAC Member (till 2025)	DST-SERB	2022
5	SIRE Fellowship (Univ of Alberta)	DST-SERB	2022
6	Institute Research Excellence Award	IIT Hyderabad	2021
7	Selected for BRICS Young Scientist Forum	BRICS	2021
8	IN YAS Member (till 2024)	IN YAS-INS A	2020
9	NASI Young Scientist Medal	INS A & NASI	2018
10	INAE Young Engineer Award	INAE	2018
11	INS A Young Scientist Medal	INS A	2017
12	Gandhian Young Technology Innovation Award	DBT-BIRAC	2017
13	Innovative Young Biotechnologist Award (IYBA)	DBT, Government of India	2016
14	Inspire Faculty Award (Biomedical)	DST, Government of India	2015
15	Institute's Award for Excellence in PhD	IIT Bombay, India	2015

Research Interests and Expertise (up to five keywords): Cancer nanotechnology, Photothermal therapy, Triggered drug delivery, Biomaterials, Nanotoxicology, Anti-Microbial Resistance (AMR) and Theranostics

Research Contribution Statistics:

ORCID ID: 0000-0003-3994-6760; **Scopus Author Identifier:** 58158015000

Description	Number	Description	Number
Journals publications	157	Books/Book chapters	35
First/Corresponding author	126	Patents – Published/filed	101
Total citation	3682	Patents - Granted	52
h-index	30	Conference abstracts/papers	20
i10-index	83	Total extramural funding as PI (Lakhs)	~ 1100
Cumulative impact factor	~ 270	Total extramural funding as Co-PI (Lakhs)	~ 2200

*Citation/h-index/ i10-index Source: Google Scholar

Selected Publications (Top 10) (* Corresponding author)

1. Appidi, T., China, D., Ştefan,**A.K.Rengan**, Wilfred Ngwa*, Engineered multifunctional nanoparticles for enhanced radiation therapy: three-in-one approach for cancer treatment. *Mol Cancer* **24**, 68 (2025). [Impact Factor: 33.9]
2. S A Sankaranarayanan.....**A.K. Rengan***, In situ thermosensitive H₂O₂/NO self-sufficient hydrogel for photothermal ferroptosis of triple-negative breast cancer, *Nanoscale* (2024)18899. [Impact factor: 5.1]
3. Gayathri Ravichandran,..... **A.K. Rengan***, The Multifaceted Role of Degradable Cobalt Nanoparticles: Dual-Target Starvation and Intracellular Acidification Engendering LC3-Associated Whole-Cell Autophagy, *ACS Materials Letters* **5** (2023) 272. [Impact Factor: 8.7]
4. Dokkari Nagalaxmi Yadav, Sri Amrutha,**A.K.Rengan***, Bioinspired Gold coated Phage Nanosomes for Anti-Microbial and Anti-CancerTheranostics *Material Today Nano* **23**(2023)2518. [Impact Factor: 8.2]
5. Syed Baseeruddin Alvi, PS Rajalakshmi,....,**A.K.Rengan*** et al. In Situ Nanotransformable Hydrogel for Chemo-Photothermal Therapy of Localized Tumors and Targeted Therapy of Highly Metastatic Tumors. *Applied Materials & Interfaces*. 2021, 13 (47), 55862-55878. [Impact Factor: 8.2]
6. PS R, Alvi SB, Begum N, Veeresh B, **Rengan A. K***. Self-Assembled Fluorosome–Polydopamine Complex for Efficient Tumor Targeting and Commingled Photodynamic/Photothermal Therapy of Triple-Negative Breast Cancer. *Biomacromolecules*. 2021 Aug 12;22(9):3926-40. [Impact Factor: 5.4]
7. Alvi SB, Rajalakshmi PS, Jogdand A, Sanjay AY, Veeresh B, John R, **Rengan A. K***. Iontophoresis mediated localized delivery of liposomal gold nanoparticles for photothermal and photodynamic therapy of acne. *Biomaterials Science*. 2021;9(4):1421-30. . [Impact Factor: 5.7]
8. Thomas A, Appidi T, Jogdand AB,....., **Rengan A. K*** et al. Facile synthesis of fluorescent polymer encapsulated metal (PoeM) nanoparticles for imaging and therapeutic applications. *ACS Applied Polymer Materials*. 2020 Feb 21;2(3):1388-97. [Impact factor: 4.7]
9. Singh SP,....**A.K.Rengan*** et al., NIR triggered liposome gold nanoparticles entrapping curcumin as in situ adjuvant for photothermal treatment of skin cancer. *International journal of biological macromolecules*. 2018, 110, 375-382. [Impact Factor: 8.5]
10. **A.K.Rengan**,....,R.Srivastava et al., In vivo analysis of biodegradable liposome gold nanoparticles as efficient agents for photothermal therapy of cancer. *Nano Letters*. 2015, 15(2), 842-848. [Impact Factor: 9.1]

PhD Students Graduated : 20; Thesis submitted : 5; Ongoing : 25;

Professional Associations: Royal Society of Chemistry-UK, Indian Association for Cancer Research (IACR), Indian National Young Academy of Sciences (INAYAS) Alumnus, State Medical Council-TN, Indian Society of Nano Medicine(ISNM), Electron Microscope Society of India.

Reviewer (Journals) : Biomaterials, ACS Applied Materials & Interfaces, Chemistry Select, ACS Biomaterials Science & Engineering, ACS Sustainable Chemistry & Engineering, Advanced Functional Materials, Bioconjugate Chemistry, Drug Delivery and Translational Research, Bioactive Materials, Journal of Controlled Release and RSC Nanoscale.

Patents (Filed/Published/Granted) :

1. A Liposomal Composition Comprising A Near Infra-Red Triimide Dye and Method of Preparation Thereof **Patent No. -554537 (Granted)**
2. A nanoparticle formulation comprising copper and omeprazole for chemodynamic therapy against skin cancer **Patent No. -562819 (Granted)**
3. A Phage-Modified Chitosan Nanocarrier and Method of Preparation Thereof **Patent No. -572484 (Granted)**
4. Porphysomal nanosystem for enhanced drug delivery and its method of preparation thereof **Patent No. 562508 (Granted)**
5. A composite Hydrogel and Method of Preparation Thereof **Patent No. -557881 (Granted)**
6. Piperine and IR820 loaded RL liposome nanoparticles for PTT/PDT of cancer and preparation thereof **Patent No. 561323 (Granted)**
7. A Nanoparticle Composition and a Method of Preparation Thereof **Patent No. -569457 (Granted)**
8. A Process for Synthesis of A Nano- Formulation (CDIR@Mnnp) **Patent No. -569061 (Granted)**
9. A modified dye and the method of preparation thereof **Patent No. -562992 (Granted)**
10. Hybrid Cell Membrane Nanosome and a Method of Preparing the Same **Patent No.- 556849 (Granted)**
11. A Biocompatible Lead-Free Material for use as an Orthopedic Implant **Patent No.- 559114 (Granted)**
12. Light-Emitting Fluorescent Molecules and A Method of Preparation Thereof **Patent No. -566735 (Granted)**
13. A Gold-Coated Bimetallic Nanocarrier Composition and Its Method of Preparation **Patent No. - 567651 (Granted)**
14. Nano-Formulation for Enhanced Therapeutic Delivery Using NIR Dye 1061 Retinoic Acid and Soy Pc **IPA No. -202441085316 (Granted)**
15. A nanoparticle formulation and a method of preparation thereof **Patent No.- 571851 (Granted)**
16. A Thermosensitive Hyogel and A Process of Preparation Thereof **Patent No. -567270 (Granted)**
17. A nanoparticle composition and a method of preparation thereof **Patent No. -556512 (Granted)**
18. Rhenium (I)-based supramolecular coordination complexes and method of preparation thereof **Patent No.: -555612 (Granted)**
19. A facial-rheniumtricarbonyl (FAC-RE(CO)₃) dinuclear metallocyclic complexes and a method of synthesis thereof **Patent No.: 554296 (Granted)**
20. Co(II) Thione Based Catalyst and Method of Preparation Thereof **Patent No. -571214 (Granted)**
21. A method for producing lipid-based nanococheles loaded with hydrophobic metformin **Patent No. - 549695 (Granted)**
22. Modified PEG-400 (mPEG-AA complex) and uses thereof **Patent No. -549413 (Granted)**
23. Chondroitin sulfate-coated nanoparticle formulation and method of preparation thereof **Patent No. - 570030 (Granted)**
24. Compositions for Prevention, Management and Treatment for Mucositis **Patent No. 541652 (Granted)**
25. High Affinity oligonucleotide nano-matrix and a nanocarrier system **Patent No. -566194 (Granted)**
26. Algosomes-Gold (Alg/Au) Complex and Method of Preparation Thereof **Patent No. -570046 (Granted)**

27. Fluorescent Polyethylene Glycol **Patent No. -438126 (Granted)**
28. A detection kit for the diagnosis of cervical cancer by quantification of visual inspection of acetic acid **Patent No. 519065 (Granted).**
29. Liponion as a multi-color fluorescent biolabelling probe. **Patent No. 362071 (Granted).**
30. Photo-disintegrable metal nanoshells for multimodal imaging and passively targeted photothermal therapy of cancer - **Patent No. -519065 (Granted).**
31. TPGS Compromised Gold Coated Poly (Lactic-co-Glycolic acid) Nanostructures and Process for its Preparation **Patent No. 354678 (Granted).**
32. Rhenium(I) Metallocycle and Method of Preparation Thereof **Patent No. 570189 (Granted).**
33. A Nanoparticle Formulation and Preparation Thereof **Patent No. 572652 (Granted).**
34. Thermosensitive hydrogel for cancer therapeutics and methods preparation thereof - **Patent No. 562333- (Granted)**
35. Degradable or Transformable gold coated liposomal nano-construct and a process for its preparation **Patent No. 437191 (Granted)**
36. Pyridine- Substituted Naphthoimidazolium compound and method of preparing the same. **Patent No. 569873 (Granted).**
37. Copper-Based Paddle wheel complex and a process of preparation thereof. **Patent No. 569830 (Granted).**
38. Thermosensitive hydrogel for cancer therapeutics and methods preparation thereof - **USP No.: 17325290 (Granted).**
39. PEG-b-ACA mixture and a method of preparing the same **Patent No. 565336 (Granted).**
40. Melanin- loaded polydopamine nanoparticles and preparation thereof **Patent No. 565009 (Granted).**
41. Barium nanocomposites for bioimaging and chemo- photothermal therapy and process for preparation thereof **Patent No. 557315 (Granted).**
42. A Formulation of Transethosomal For Targeted Drug Delivery And a Method Of Preparing The Same **Patent No. 572660 (Granted).**
43. A Fluid Collection Device for Collecting Fluids **Patent No. 572356 (Granted).**
44. Phage modified calcium peroxide nanoparticles and a method of preparation thereof **Patent No. 563555 (Granted).**
45. Encapsulated nano-formulations of Buparvaquone and methods of preparation thereof **Patent No. 564402 (Granted).**
46. Liposome- Encapsulated Neem Nanoemulsion and a Method of Preparation Thereof **Patent No. 568225 (Granted).**
47. FL-Peg Nanoparticles Encapsulating UGS for Cancer Therapy **Patent No. 569148 (Granted).**
48. Emulsomes for Melanoma Cancer Treatment and Method of Preparation Thereof **Patent No. 569926 (Granted).**
49. Copper (Cu(Ii)) Complex and A Method of Preparation Thereof **Patent No. 570036 (Granted).**
50. Multi-Functional Lipid Nanoemulsions for Dual Therapy of Infectious Diabetic Wounds using Quercetin, CE6, and Rosemary Oil **Patent No. 565406 (Granted).**

51. Self-Assembled Nanoparticle Formulation for Anti-Cancer Activity and Methods of Preparation Thereof **Patent No. 569710 (Granted).**
52. Phage nanosomes for NIR dye delivery and methods of preparation thereof **IPA No.: 202441036196**
53. Modified PEG-400 (mPEG-AA complex) and uses thereof - **USP No.: 17403964**
54. A Flavin Analogue for Lysosome Tracking and Photodynamic Therapy **IPA No.- 202441000773**
55. A Pharmaceutical Composition and a Method of Preparation Thereof **IPA No. -202441036487**
56. A Wound Dressing for Preventing Biofilm formation on a Wound **IPA No. -202441036378**
57. Chitosan-Conjugated Carboxymethyl Inulin Modified Gold Nanoclusters Incorporating IR-820 and DNA-Conjugated pphs for Biomedical Application and Preparation Thereof **IPA No. -202441035158**
58. Liposomal Nanoparticles Encapsulating Boldine and Ir- 775 for Therapy Against Metastatic Triple-Negative Breast Cancer and Preparation Thereof **IPA No. -202441034692**
59. A Phage-Modified Casein Nanocarrier Conjugated With Glutathione and Method of Preparation Thereof **IPA No. -202441080547**
60. Pyridine-Substituted Naphthoimidazolium Compound and Method of Preparing The Same **IPA No. - 202441084005**
61. Composition for Delivery of Therapeutic Complex Within Golgi Apparatus and Method of Producing The Same **IPA No. -202441084476**
62. Liposome of Saussurea Obvallata and Its Process of Preparation **IPA No. -202441086405**
63. A Liposome Comprising Bronzaphyrin and Method of Preparation Thereof **IPA No. -202441086634**
64. A Nanoparticle Composition and a Method of Preparation Thereof **IPA No. -202441086473**
65. A Functionalized Complex Nanoparticle Composition and a Method of Preparation Thereof **IPA No. - 202441086474**
66. A Pharmaceutical Composition and a Method of Preparation Thereof **IPA No. -202441085921**
67. Supramolecular Hydrogels and Method of Preparation Thereof **IPA No. -202441086354**
68. A Nanoliposomal Formulation and a Method of Preparation Thereof **IPA No. -202441086472**
69. A Liposomal Formulation and Preparation Thereof **IPA No. -202441085475**
70. A Nanoparticle Composition Having Anti-Cancer Activity and A Method of Preparation Thereof **IPA No. - 202441082537**
71. Gallic Acid and Ir-775 Loaded Liposome Nanoparticles for Phototherapy of Cancer and Preparation Thereof **IPA No. -202441084440**
72. A Nanoliposome Formulation to Enhance The Efficacy of Doxorubicin in Prostate Cancer **IPA No. - 202441090417**
73. Solvent-Based Synthesis of Gold Nanostructures for Enhanced Anticancer Efficacy **IPA No. -202441091408**
74. A Formulation Comprising Iohexol Functionalized Dopamine Compound and a Method of Preparation Thereof **IPA No. -202441102596**
75. Fluorescent Enhanced Silver Triflate Nanoclusters Using Silver Triflate Precursor **IPA No. -202441101142**
76. A Membrane Cloaked Nanosome **IPA No. -202441036003**
77. Co-Crystal Of Copper Iodide (Cui) Clusters and a Process for Preparation Thereof **IPA No. -202541003577**

78. Resveratrol and coumarin 6 loaded nanoparticle formulation for cancer therapy and preparation thereof **IPA No. -202541009119**
79. A ruthenium-based thione complex having anticancer activity and a method of preparation thereof **IPA No. -202541021105**
80. A ruthenium-based thione complex having anticancer activity and a method of preparation thereof **IPA No. -202541021105**
81. Ferrocene-coumarin complex and a method of preparation thereof **IPA No. -202541021928**
82. Thulium Containing Caffeic Acid-Terpyridine Mof And A Method Of Preparation Thereof **IPA No. -202541021928**
83. A Ferro-Flexible Antibacterial Composite Film And Method Of Preparation Thereof **IPA No. -202541021928**
84. Drug-Loaded Soya Pc Nanoplatfrom And Process Of Preparing The Same **IPA No. -202541021928**
85. A Mn-Based Nanocarrier Formulation And A Process For Preparation Thereof **IPA No. -202541021928**
86. A Cyclodextrin- Isatin Nanoparticle Formulation And A Method Of Preparation Thereof **IPA No. -202541021928**
87. A Fluorescent Zinc-Based Nanocluster And A Method For Preparation Thereof **IPA No. -202541021928**
88. Fluorescent Antibacterial Formulation, And Method Of Preparation Thereof **IPA No. -202541021928**
89. Silver-Coated Nanoparticle Formulation And A Method Of Preparation Thereof **IPA No. -202541021928**
90. Liposome Nanoparticle Formulation And A Method Of Preparation Thereof **IPA No. -202541021928**
91. A Method For Synthesizing Gold Coated Calcium Peroxide Nano Particles **IPA No. -202541021928**
92. Bioinspired gold coated phage nanosomes and methods of preparation therefor **IPA No. 202341013944**
93. A multimodal liposomal composition for Naja Naja Venom neutralization and a method for producing the same - **IPA No. 202241024566.**
94. Hybrid Lipo-Polymeric Nanoparticles and Methods of Preparation Thereof- **IPA No. 202141016474**
95. Plant-derived fluorescent nanoparticles for treatment of cancer- **IPA No. 201941030649.**
96. Ferroelectric polymer (β -pvdf) for control and mitigation of Microbes under small voltage signals - **IPA No. 202041050666.**
97. Hybrid polymeric nanoformulation for the treatment and management of Retinoblastoma **IPA No. 2141/CHE/2020**
98. Fluorescent Polymeric NPs as imaging probe- **IPA No. 032865/CHE/2018**
99. Liposomal nanoformulation for treatment of cancer **IPA No. 35668 /CHE/2018**
100. Enzymatically degradable Lipos Au Nanoparticles for Cancer Theranostics - **IPA No. 4910/MUM/2015.**
101. Modified PEG-400 (mPEG-AA complex) and uses thereof. **IPA No. 4134/CHE/2021**

Review Papers

1. Srivastava, R.; Eswar, K.; Ramesh, S.S.R.; Prajapati, A.; Sonpipare, T.; Basa, A.; Gubige, M.; Ponnappalli, S.; Shashidhar, S.; Rengan, A.K., Zebrafish as a Versatile Model Organism: From Tanks to Treatment, 2025, MedComm - Future Medicine, John Wiley and Sons Inc
2. Gedda, G.; Putta, C.L.; Verma, A.; Ramesh, S.S.R.; Girma, W.M.; Park, Y.-J.; Rengan, A.K.; Pang, M.-G., Carbon dots: A promising tool for viral infection treatment, 2025, Materials Today Bio, Elsevier B.V.
3. Mukherjee, S.; Rengan, A.K.; Patra, C.R., Advances in Nanomedicine for Cancer Theranostics, 2025, Biomedical Materials (Bristol), Institute of Physics
4. Chopra, P.; Fatima, A.; Mohapatra, S.; Murugaiyan, K.; Vemuganti, G.K.; Rengan, A.K.; Watson, S.L.; Singh, V.; Basu, S.; Singh, S., Extracellular vesicles in dry eye disease and Sjögren's syndrome: A systematic review on their diagnostic and therapeutic role, 2025, Survey of Ophthalmology, Elsevier Inc.
5. Srivastava, A.; Joshi, M.; Rengan, A.K., Feeding the future: the role of nanotechnology in tailored nutrition, 2025, Nucleus (India), Springer
6. Mukherjee, S.; Verma, A.; Kong, L.; Rengan, A.K.; Cahill, D.M., Advancements in Green Nanoparticle Technology: Focusing on the Treatment of Clinical Phytopathogens, 2024, Biomolecules, Multidisciplinary Digital Publishing Institute (MDPI)
7. Buddhiraju, H.S.; Yadav, D.N.; Dey, S.; Eswar, K.; Padmakumar, A.; Rengan, A.K., Advances in Peptide-Decorated Targeted Drug Delivery: Exploring Therapeutic Potential and Nanocarrier Strategies, 2024, ACS Applied Bio Materials, American Chemical Society
8. Saha, S.; Trivedi, D.; Rengan, A.K.; Yu, A., Polyurethane Nanoparticles as Versatile Tools in Nanomedicine: A Review, 2024, ACS Applied Nano Materials, American Chemical Society
9. Sharma, K.; Dey, S.; Karmakar, R.; Rengan, A.K., A comprehensive review of 3D cancer models for drug screening and translational research, 2024, Cancer Innovation, John Wiley and Sons Inc
10. Beniwal, N.; Verma, A.; Putta, C.L.; Rengan, A.K., Recent Trends in Bio-nanomaterials and Non-invasive Combinatorial Approaches of Photothermal Therapy against Cancer, 2024, Nanotheranostics, Ivyspring International Publisher
11. Pebam, M.; Buddhiraju, H.S.; Verma, A.; Rengan, A.K., Nano Phyto-Phenols as potential immunomodulators: A review, 2023, Food and Humanity, Elsevier B.V.
12. Chowdary, P.; Padmakumar, A.; Rengan, A.K., Exploring the potential of transthesosomes in therapeutic delivery: A comprehensive review, 2023, MedComm - Biomaterials and Applications, John Wiley and Sons Inc
13. Mascarenhas-Melo, F.; Mathur, A.; Murugappan, S.; Sharma, A.; Tanwar, K.; Dua, K.; Singh, S.K.; Mazzola, P.G.; Yadav, D.N.; Rengan, A.K.; Veiga, F.; Paiva-Santos, A.C., Inorganic nanoparticles in dermatopharmaceutical and cosmetic products: Properties, formulation development, toxicity, and regulatory issues, 2023, European Journal of Pharmaceutics and Biopharmaceutics, Elsevier B.V.
14. Karmakar, R.; Dey, S.; Alam, A.; Khandelwal, M.; Pati, F.; Rengan, A.K., Attributes of Nanomaterials and Nanotopographies for Improved Bone Tissue Engineering and Regeneration, 2023, ACS Applied Bio Materials, American Chemical Society
15. Ali, S.; Hak, A.; Pogu, S.V.; Rengan, A.K., Radiotherapy, photodynamic therapy, and cryoablation-induced abscopal effect: Challenges and future prospects, 2023, Cancer Innovation, John Wiley and Sons Inc

16. Putta, C.L.; Eswar, K.; Rengan, A.K., Melatonin: Avenues in cancer therapy and its nanotechnological advancements, 2023, MedComm - Biomaterials and Applications, John Wiley and Sons Inc
17. Alam, A.; Karmakar, R.; Rengan, A.K.; Khandelwal, M., Nanofiber-Based Systems for Stimuli-Responsive and Dual Drug Delivery: Present Scenario and the Way Forward, 2023, ACS Biomaterials Science and Engineering, American Chemical Society
18. Padmakumar, A.; Chowdary, P.; Eswar, K.; Kong, L.; Yang, W.; Gopalakrishnan, S.; Cahill, D.M.; Rengan, A.K., Bacteria-Premised Nanobiopesticides for the Management of Phytopathogens and Pests, 2023, ACS Agricultural Science and Technology, American Chemical Society
19. Eswar, K.; Mukherjee, S.; Prabusankar, P.; Rengan, A.K., Immunomodulatory natural polysaccharides: An overview of the mechanisms involved, 2023, European Polymer Journal, Elsevier Ltd
20. Pogu, S.V.; Dehariya, D.; Yadav, D.N.; Rengan, A.K., A review on fabrication, actuation, and application of magnetic force driven, light driven and DNA nano/microrobots in modern theranostics, 2023, Molecular Systems Design and Engineering, Royal Society of Chemistry
21. Khatun, S.; Putta, C.L.; Hak, A.; Rengan, A.K., Immunomodulatory nanosystems: An emerging strategy to combat viral infections, 2023, Biomaterials and Biosystems, Elsevier Ltd
22. Hak, A.; Ali, S.; Sankaranarayanan, S.A.; Shinde, V.R.; Rengan, A.K., Chlorin e6: A Promising Photosensitizer in Photo-Based Cancer Nanomedicine, 2023, ACS Applied Bio Materials, American Chemical Society
23. Yadav, D.N.; Ali, S.; Thanekar, A.M.; Pogu, S.V.; Rengan, A.K., Recent Advancements in the Design of Nanodelivery Systems of siRNA for Cancer Therapy, 2022, Molecular Pharmaceutics, American Chemical Society
24. Khatun, S.; Appidi, T.; Rengan, A.K., Casein nanoformulations - Potential biomaterials in theranostics, 2022, Food Bioscience, Elsevier Ltd
25. Chinchulkar, S.A.; Patra, P.; Dehariya, D.; Yu, A.; Rengan, A.K., Polydopamine nanocomposites and their biomedical applications: A review, 2022, Polymers for Advanced Technologies, John Wiley and Sons Ltd
26. Patra, P.; Rengan, A.K., Cancer cell membrane cloaked nanocarriers: A biomimetic approach towards cancer theranostics, 2022, Materials Today Communications, Elsevier Ltd
27. Koyande, N.P.; Srivastava, R.; Padmakumar, A.; Rengan, A.K., Advances in Nanotechnology for Cancer Immunoprevention and Immunotherapy: A Review, 2022, Vaccines, MDPI
28. Ravichandran, G.; Yadav, D.N.; Murugappan, S.; Sankaranarayanan, S.A.; Revi, N.; Rengan, A.K., "Nano effects": a review on nanoparticle-induced multifarious systemic effects on cancer theranostic applications, 2022, Materials Advances, Royal Society of Chemistry
29. Pebam, M.; Sushma, M.V.; Sankaranarayanan, S.A.; Thanekar, A.M.; Koyande, N.P.; Rengan, A.K., Antiviral perspectives of economically important Indian medicinal plants and spices, 2022, Proceedings of the Indian National Science Academy, Springer Nature
30. Pratyusha, P.; Gopalakrishnan, S.; Pebam, M.; Rengan, A.K., Nano-biofertilizers on soil health, chemistry, and microbial community: benefits and risks, 2022, Proceedings of the Indian National Science Academy, Springer Nature
31. Shinde, V.R.; Revi, N.; Murugappan, S.; Singh, S.P.; Rengan, A.K., Enhanced permeability and retention effect: A key facilitator for solid tumor targeting by nanoparticles, 2022, Photodiagnosis and Photodynamic Therapy, Elsevier B.V.

32. Padmakumar, A.; Koyande, N.P.; Rengan, A.K., The Role of Hitchhiking in Cancer Therapeutics—A Review, 2022, Advanced Therapeutics, John Wiley and Sons Inc
33. Koyande, N.P.; Gangopadhyay, M.; Shashidhar, S.; Rengan, A.K., The role of gut microbiota in the development of colorectal cancer: a review, 2022, International Journal of Colorectal Disease, Springer Science and Business Media Deutschland GmbH
34. Sankaranarayanan, S.A.; Thomas, A.; Revi, N.; Ramakrishna, B.; Rengan, A.K., Iron oxide nanoparticles for theranostic applications - Recent advances, 2022, Journal of Drug Delivery Science and Technology, Editions de Sante
35. Khatun, S.; Appidi, T.; Rengan, A.K., The role played by bacterial infections in the onset and metastasis of cancer, 2021, Current Research in Microbial Sciences, Elsevier Ltd
36. Revi, N.; Rengan, A.K., Impact of dietary polyphenols on neuroinflammation-associated disorders, 2021, Neurological Sciences, Springer-Verlag Italia s.r.l.
37. Hak, A.; Ravasaheb Shinde, V.; Rengan, A.K., A review of advanced nanoformulations in phototherapy for cancer therapeutics, 2021, Photodiagnosis and Photodynamic Therapy, Elsevier B.V.
38. Thanekar, A.M.; Sankaranarayanan, S.A.; Rengan, A.K., Role of nano-sensitizers in radiation therapy of metastatic tumors, 2021, Cancer Treatment and Research Communications, Elsevier Ltd
39. Ravichandran, G.; Rengan, A.K., Aptamer-mediated nanotheranostics for cancer treatment: A review, 2020, ACS Applied Nano Materials, American Chemical Society
40. Darabdhara, G.; Das, M.R.; Singh, S.P.; Rengan, A.K.; Szunerits, S.; Boukherroub, R., Ag and Au nanoparticles/reduced graphene oxide composite materials: Synthesis and application in diagnostics and therapeutics, 2019, Advances in Colloid and Interface Science, Elsevier B.V.
41. Yadav, P.; Singh, S.P.; Rengan, A.K.; Shanavas, A.; Srivastava, R., Gold laced bio-macromolecules for theranostic application, 2018, International Journal of Biological Macromolecules, Elsevier B.V.

Book Chapters

1. Eswar, K.; Yadav, D.N.; Rengan, A.K., Silica-based nanoparticles in biosensors, 2025, Elsevier
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